



Rupankar Ganguly

Final-year B.Tech, Computer Science & Engineering (2027 Batch)

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SKILLS

Languages: Java , Python, JavaScript, SQL
Backend : Spring Boot, FastAPI, REST API Design, JWT Authentication, OOP
Frontend: React, HTML5, CSS3, Tailwind CSS
Databases: PostgreSQL, MySQL, MongoDB Atlas.
AI / GenAI: LangChain, LangGraph, RAG, Embeddings, Groq LLM API, Agentic Tool-Calling, Prompt Engineering.
Tools & Cloud: Git/GitHub , Postman, Streamlit, AWS (Cloud Practitioner), CCNA networking basics .
Core CS: Data Structures & Algorithms, OOP, SQL .
Cloud: AWS Cloud Fundamentals, EC2, VPC, RDS, Lambda, CloudWatch, AWS Security & Pricing

DATA STRUCTURES & ALGORITHMS

- Solved **150+ problems** on LeetCode **max contest rating 1480** (profile: leetcode.com/u/rupankarganguly492).
- Strongest topics: arrays, hashing, trees, graphs, DP .

PROJECTS

Faculty Review System (login system was added specifically to demonstrate security implementation skills) / [Live Demo](#)
FastAPI, PostgreSQL, JWT, JavaScript

- Designed **8 active REST APIs** for review submission and retrieval with a focus on clean framework integration and service reliability.
- Added **JWT-based authentication**, email verification, token verification, and ownership checks across **5 frontend pages** to prevent duplicate or fraudulent reviews and strengthen maintenance of platform integrity.

AI Task Manager Agent / *LangChain, LangGraph, Groq LLM API, Streamlit* [Live Demo](#)

- Built a tool-calling AI agent that manages a to-do database through natural language commands.
- Designed **4 database-backed** tools (create, list, update, delete) with status/priority mapping, turning phrases like “mark as done” into state changes.
- Used the LangGraph checkpointer for persistent multi-turn memory tied to a session thread ID.

RAG Document Q&A Bot / *LangChain, Gemini Embeddings, Groq LLM, Streamlit* [Live Demo](#)

- Built an agentic RAG assistant for PDF Q&A: loads PDFs, splits them into **1000-character** chunks (**200 overlap**), and embeds them with Google Gemini embeddings into a LangChain in-memory vector store.
- A LangChain agent (**Groq LLM**) calls a retriever tool for **top-4 similarity** search and answers only from the retrieved context, replying **“I don’t know”** when the answer is not in the document; per-session chat memory via the **LangGraph checkpointer**.

Movie Recommendation System / *Spring Boot, MongoDB Atlas, React* [Live Demo](#)

- Built the Spring Boot backend for a full-stack movie recommendation platform in a team; REST APIs are consumed by a teammate’s React frontend.
- Modeled MongoDB Atlas collections for user preferences and movie metadata to support personalized recommendations.

EXPERIENCE

Virtual Internship (Cisco + AICTE), Cisco Networking Academy / *Remote* May 2026 – Jul 2026

- Architected** NetSage AI as the internship capstone in a **2-3 member** team.
- NetSage AI uses symptom reports to diagnose wireless, **VLAN, DHCP, DNS**, routing, **ACL, NAT**, and other network faults. show-command evidence, improving service triage.
- Used **33 test cases** from **eight fault categories** to validate the dashboard while a teammate used the AI diagnosis and prompt layer.

EDUCATION

B.Tech in Computer Science & Engineering Expected Jul 2027

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar | CGPA: 7.37

CERTIFICATIONS

- AWS Cloud Practitioner Essentials** – AWS Training & Certification
- Generative AI: Introduction and Applications** – IBM (Coursera)
- Cisco certificate** – SQL, Python.
- Cisco CCNA** – Introduction to Networks; Switching, Routing & Wireless Essentials; Enterprise Networking, Security & Automation